Safety-Oriented Design of Cycling Infrastructure Minimizing Impact on Vehicular Traffic

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***Abstract*— Designing safer cycling infrastructure without causing major traffic delays is a key challenge in city planning. This paper presents a bi-level optimization framework to address and solve this problem. The upper level plans bike lane upgrades to reduce cyclists' exposure to traffic, while ensuring car travel times remain reasonable despite potential impacts on road capacity. The lower level models and determines cyclists' route choices based on factors such as slope, turns, and traffic conditions on bike paths. We present a case study for Grenoble network, France, and a safety-oriented upgrade of the existing cycling infrastructure is determined using a genetic algorithm. To understand how cars are affected by such a modification of the cycling infrastructure, we simulate how drivers might change their routes after bike lanes upgrades. Our results show that cyclists' safety can be improved with only small impacts on car travel times, thus highlighting the potential of this approach to support safer, more sustainable urban mobility.**

***Index Terms*— Cycling infrastructure, bi-level optimization, safety, urban planning, traffic assignment.**

I. Introduction

Cycling brings substantial benefits to health, the environment, and the economy by lowering emissions, reducing travel costs, and promoting public health [1]. Consequently, it is increasingly promoted as a sustainable mode for short trips. Yet, safety and comfort concerns remain major barriers to broader adoption [2]. Research shows that physically separated bike lanes improve perceived safety and encourage cycling [3], and that well-connected infrastructure further promotes usage [4]. However, reallocating road space for bikes may reduce car lane capacity, raising concerns about congestion. Given limited budgets, cities must strategically invest to enhance cyclist safety and comfort while minimizing impacts on car travel. This paper presents a framework to support such decisions.

Related Work: Cycling infrastructure planning has been approached from multiple angles. Some works focus on selecting links to upgrade to improve bike connectivity while limiting car travel time increases [5], [6]. The framework in [7] balances bike and car travel times through multi-criteria optimization. Other approaches, such as [8], prioritize high-demand corridors while limiting car traffic disruption. Similarly, [9] uses heuristics to design cost-effective bike networks under budget limits, minimizing car impacts.

Studies like [10], [11] aim to maximize the societal benefit-cost ratio of cycling infrastructure, focusing on connectivity and feasibility. However, many omit direct safety considerations or model both modes with travel time as the only objective. This paper fills that gap by integrating safety and traffic dynamics into infrastructure planning—a combination not previously addressed.

Organization: Section II describes data sources, bike lane classification, and the overall modeling framework. Section III details the solution algorithm. Section IV presents the Grenoble case study. Section V concludes with findings and future directions.

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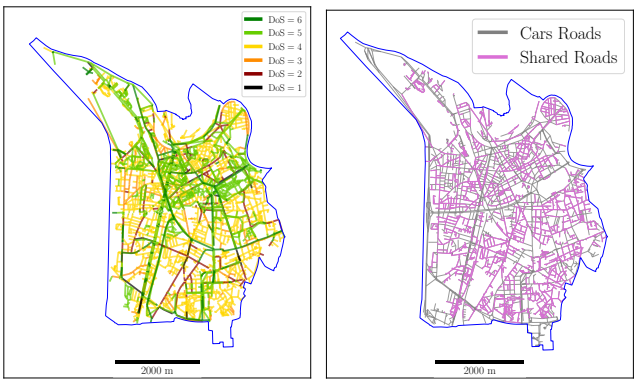


Fig. 1: Multi-modal transportation network of Grenoble, France. *Left:* Cycling infrastructure classified by degree of separation (DoS) of bike lanes, ranging from 1 (least separated) to 6 (fully separated). *Right:* Road network for cars, including links shared with bicycles.

DoS	Description	AADT (veh/day)
6	Dedicated bike lanes, fully separated from motor traffic.	14 400
5	Bike lanes shared with pedestrians, separate from cars.	16 000
4	Mixed-traffic roads (30 km/h) without designated bike lanes.	960
3	Shared roads with cycling pictograms.	3 120
2	Roads with painted bike lanes.	6 912
1	Unmarked shared car-bike lanes.	8 448

TABLE I: The degree of separation (DoS) categorizes bike lane infrastructure based on safety levels. The third column presents annual average daily traffic (AADT) on adjacent car roads, sourced from the GTLVille platform in Grenoble [12].

II. Model Design

This section introduces the bike lane classification, the graph-based network model, the cyclist flow prediction, car travel time modeling, and the bi-level optimization that jointly minimizes cyclist exposure and car travel time by steering bike flows toward safer routes.

A. Bike Lane Safety Classification

We adopt the DoS classification from an OSM-based platform to evaluate cycling infrastructure based on separation from motor traffic. Table I outlines the DoS levels, where higher values indicate safer lanes. In our model, DoS serves as the weight in the bicycle sub-network graph. An upgrade corresponds to moving from a lower to a higher DoS. The classification approach is detailed in [13], [14]. Figure 1 shows both the DoS map and the car network in Grenoble.

B. Graph Representation

The multimodal transport system is modeled as a weighted digraph $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$, with nodes $\mathcal{V}$, directed arcs $\mathcal{A}$, and weights $\mathcal{W}$. It consists of two overlapping layers: bicycle $\mathcal{G}^b = (\mathcal{V}^b, \mathcal{A}^b, \mathcal{W}^b)$ and car $\mathcal{G}^c = (\mathcal{V}^c, \mathcal{A}^c, \mathcal{W}^c)$. Each arc $a \in \mathcal{A}^b$ has weight w_a^b equal to its DoS and is symmetric; car arcs use uniform weight $w_a^c = 1$. Geographical lengths are d_a^b and d_a^c. Layers may overlap in nodes and arcs, i.e., $\mathcal{V}^b \cap \mathcal{V}^c \neq \emptyset$ and $\mathcal{A}^b \cap \mathcal{A}^c \neq \emptyset$. We use arc notation a across both layers, with superscripts b or c as needed. Origins and destinations are denoted $\mathcal{V}_{\mathcal{O}}^b$, $\mathcal{V}_{\mathcal{D}}^b$ for bikes, and similarly for cars.

C. Cyclist Exposure

Let $q_{(u,v)}^b$ be the cycling demand between origin-destination pair $(u,v) \in \mathcal{V}_{\mathcal{O}}^b \times \mathcal{V}_{\mathcal{D}}^b$. Following [5], we apply an all-or-nothing assignment, allocating demand to the shortest path based on a cyclist-specific distance metric. This metric incorporates infrastructure topology and reflects perceived difficulty and unsafe. Using betweenness centrality [15], flow is assigned along shortest paths.

For each arc $a \in \mathcal{A}^b$, we define: d_a^b: length, $\measuredangle_a^b$: angular change at intersections, s_a^b: slope adjustment from terrain gradient m_a^b, f_a^b: traffic factor due to adjacent car flows.

The cyclist distance $\bar{d}_{(u,v)}^b$ is:

$$\bar{d}_{(u,v)}^b = \sum_{a \in p_{(u,v)}^b} \left(d_a^b \cdot s_a^b \cdot (f_a^b)^\rho \right) + \sum_{a \in p_{(u,v)}^b} \measuredangle_a^b \cdot \kappa, \quad (1)$$

with parameters $\rho, \kappa > 0$.

The traffic factor f_a^b depends on DoS and AADT:

$$f_a^b = \begin{cases} \psi \cdot \exp(\text{AADT}(a)) & \text{if } w_a^b < 5, \\ \psi \cdot 2.03 & \text{if } w_a^b = 5, \\ \psi & \text{if } w_a^b = 6, \end{cases} \quad (2)$$

where ψ is a scaling constant. As per [16], AADT effects vanish for DoS 5–6. The 2.03 factor reflects the higher crash risk in shared pedestrian spaces [17].

Let $p_{(u,v)}^b$ be the shortest path under Eq. (1), with flow $\phi_{p_{(u,v)}^b}^b = q_{(u,v)}^b$. Define a binary indicator $\mathbb{I}_{a \in p_{(u,v)}^b}$ as 1 if arc a is in path $p_{(u,v)}^b$. Then, total arc flow is:

$$\phi_a^b = \sum_{(u,v)} q_{(u,v)}^b \cdot \mathbb{I}_{a \in p_{(u,v)}^b} \geq 0. \quad (3)$$

Cyclist exposure $\mathcal{E}$ on the network is the sum of arc-level exposures:

$$\mathcal{E} = \sum_{a \in \mathcal{A}^b} \phi_a^b \cdot f_a^b \cdot d_a^b, \quad (4)$$

indicating that exposure increases with higher flows, greater traffic influence, and longer travel distance.

D. Cars Travel Time

Similarly to the bicycle network, let $q_{(u,v)}^c$ denote the demand between nodes $(u,v) \in \mathcal{V}_{\mathcal{O}}^c \times \mathcal{V}_{\mathcal{D}}^c$. Unlike the all-or-nothing assignment, we adopt the UE traffic assignment principle [18], which assumes that no driver can reduce their travel time by unilaterally changing routes. Capturing this equilibrium requires accounting for capacity constraints: as demand increases or capacity decreases, travel times on a link increase, possibly making alternative paths more appealing. Let ϕ_a^c denote the car flow on arc $a \in \mathcal{A}^c$, and $t_a^c(\phi_a^c)$ be the corresponding travel time. We use the Bureau of Public Roads (BPR) function [19]:

$$t_a^c(\phi_a^c) = t_a^0 \left(1 + \mu \left(\frac{\phi_a^c}{\theta_a^c} \right)^\sigma \right), \quad (5)$$

where t_a^0 is the free-flow travel time, θ_a^c is the arc capacity, and $\mu = 0.15$, $\sigma = 4$ are fixed parameters. To compute equilibrium flows, let $\mathcal{P}_{(u,v)}^c$ be the set of car paths between each $(u,v) \in \mathcal{V}_{\mathcal{O}}^c \times \mathcal{V}_{\mathcal{D}}^c$, and $\phi_{(u,v)}^{p^c}$ the path flow on $p^c \in \mathcal{P}_{(u,v)}^c$. The demand constraint reads:

$$\sum_{p^c \in \mathcal{P}_{(u,v)}^c} \phi_{(u,v)}^{p^c} = q_{(u,v)}^c. \quad (6)$$

Total arc flow is obtained by summing over path flows:

$$\phi_a^c = \sum_{(u,v)} \sum_{p^c \in \mathcal{P}_{(u,v)}^c} \phi_{(u,v)}^{p^c} \cdot \mathbb{I}_{a \in p^c}, \quad (7)$$

where $\mathbb{I}_{a \in p^c} = 1$ if arc a is part of path p^c, and 0 otherwise. Flows must be non-negative:

$$\phi_{(u,v)}^{p^c} \geq 0, \quad \forall p^c, \forall (u,v), \quad (8)$$

$$\phi_a^c \geq 0, \quad \forall a \in \mathcal{A}^c. \quad (9)$$

The total travel time is defined as the sum over all links of the product of flow and travel time using [20]:

$$\mathcal{TTT} = \sum_{a \in \mathcal{A}^c} \phi_a^c \cdot t_a^c(\phi_a^c), \quad (10)$$

where $\boldsymbol{\phi}^c = [\phi_a^c]_{a \in \mathcal{A}^c}^\top \in \mathbb{R}_{\geq 0}^{|\mathcal{A}^c|}$. The UE traffic assignment can thus be formulated as:

Problem 1 (UE Traffic Assignment):

$$\boldsymbol{\phi}^{c^\star} \in \mathbb{R}_{\geq 0}^{|\mathcal{A}^c|} = \arg \min_{\boldsymbol{\phi}^c \in \mathbb{R}_{\geq 0}^{|\mathcal{A}^c|}} \sum_{a \in \mathcal{A}^c} \int_0^{\phi_a^c} t_a^c(x)\, \mathrm{d}x$$

s.t. (6) – (9).

Then, $\boldsymbol{\phi}^{c^\star} = \left[\phi_a^{c^\star}\right]_{a \in \mathcal{A}^c}^\top$ represents the UE flow assignment, in which no driver can unilaterally reduce their travel time. This flow pattern satisfies the conditions of a Nash equilibrium, as defined in the classical traffic assignment model.

The convexity of the total travel time functional $\mathcal{TTT}$ and its connection to user equilibrium are formally established in [21]. In the next section, we integrate this car traffic assignment model with the cyclist exposure model to form the full bi-level optimization framework.

E. Interaction Between Cars and New Bike Lanes

Upgrading bike lanes can reduce the capacity of adjacent car lanes. However, as shown in [22], this effect is negligible for bike lanes with a DoS ≤ 4. In our setting, only fully separated bike lanes, i.e., those with $\tilde{w}_a^b \geq 5$, affect car capacity. The capacity reduction is modeled as:

$$\tilde{\theta}_a^c = \frac{1}{2}\theta_a^c, \quad \text{if } \tilde{w}_a^b \geq 5,\, a \in \mathcal{A}^c \cap \mathcal{A}^b, \tag{11}$$

where $\tilde{\theta}_a^c$ is the updated car capacity on arc a, θ_a^c is the original capacity, and $\tilde{w}_a^b$ is the DoS of the upgraded bike lane on the shared arc. Following [7], the $\frac{1}{2}$ factor reflects that a standard bike lane occupies roughly half the width of a car lane, thus reducing vehicular capacity by 50%. We assume that drivers do not adjust their routes in response to these changes. Therefore, the car flow vector $\phi^{c^\star}$ obtained from Problem 1—computed using the original capacities—remains fixed and represents the pre-upgrade UE traffic assignment, serving as the baseline for our main optimization problem. However, the updated capacities $\tilde{\theta}_a^c$ in Eq. (5) may still lead to increased travel times. We now introduce the bi-level optimization framework, which integrates the cyclist exposure model with this fixed-flow assumption, while capturing the impact of fully separated bike lane upgrades on car travel times.

F. Optimization Problem

Given a fixed budget B, the objective is to upgrade the cycling infrastructure by increasing the DoS on selected segments, minimizing cyclist exposure while limiting car travel time increases. Car flows $\phi^{c^\star}$ remain fixed during optimization, obtained with the pre-upgrade UE assignment from Problem 1. However, travel time penalties from high-DoS upgrades are accounted for via updated capacities following Eq. (11). Each upgrade incurs a cost and must comply with a global budget constraint.

- $\mathbf{w}^b = \left[w_a^b\right]_{a\in\mathcal{A}^b}^\top \in \mathbb{N}^{|\mathcal{A}^b|}$: vector of original DoS values for each arc in the bike network;
- $\tilde{\mathbf{w}}^b = \left[\tilde{w}_a^b\right]_{a\in\mathcal{A}^b}^\top \in \Omega$: vector of decision variables (candidate configuration) representing the upgraded DoS values, where Ω is the constraint set that defined as $\Omega = \left\{\tilde{w}^b \in \mathbb{N}^{|\mathcal{A}^b|} \colon w_a^b \leq \tilde{w}_a^b \leq 6, \quad \forall a \in \mathcal{A}^b\right\}$;

Additionally, we define the following variables, all normalized between their respective minimum and maximum values:

Now, we present the optimization problem that consists of two interconnected levels:

Symbol	Description	Domain	Unit
$\phi_a^b(\tilde{\mathbf{w}}^b)$	Flow on arc $a \in \mathcal{A}^b$	$\mathbb{R}_{\geq 0}$	bike/hr
d_a^b	Length of $a \in \mathcal{A}^b$	$\mathbb{R}_{\geq 0}$	m
$\bar{d}_a^b$	Cyclist distance of $a \in \mathcal{A}^b$	$\mathbb{R}_{\geq 0}$	m
$\mathcal{E}_a(\tilde{\mathbf{w}}^b)$	Cyclist exposure on $a \in \mathcal{A}^b$	$\mathbb{R}_{\geq 0}$	-
$\tilde{\theta}_a^c(\tilde{\mathbf{w}}^b)$	Updated car lane capacity on $a \in \mathcal{A}^c$	$\mathbb{R}_{\geq 0}$	veh/hr
$\phi_a^{c^\star}$	Pre-upgrade car flow on lane $a \in \mathcal{A}^c$ obtained by solving Problem 1	$\mathbb{R}_{\geq 0}$	veh/hr
$\pi(w_a^b, \tilde{w}_a^b)$	Upgrade cost for $a \in \mathcal{A}^b$	$\mathbb{R}_{\geq 0}$	\$

TABLE II: Abbreviated notation for variables in cycling network model, writing $(\tilde{\mathbf{w}}^b)$ next to a variable indicates that it is a function of the candidate configuration for bike lane improvements.

Problem 2 (Trade-Off Network-Upgrade):

$\mathbb{U}_{\text{pper}}$:

$$\mathbf{w}^\star = \arg\min_{\tilde{\mathbf{w}}^b \in \Omega} \; \epsilon \cdot \left(\sum_{a\in\mathcal{A}^b} \phi_a^{b^\star}(\tilde{\mathbf{w}}^b) \cdot f_a^b(\tilde{\mathbf{w}}^b) \cdot d_a^b \right) + (1-\epsilon) \cdot \left(\sum_{a\in\mathcal{A}^c} \phi_a^{c^\star} \cdot t_a^0 \cdot \left(1 + \mu \left(\frac{\phi_a^{c^\star}(\tilde{\mathbf{w}}^b)}{\tilde{\theta}_a^c(\tilde{\mathbf{w}}^b)} \right)^\sigma \right) \right) \tag{12}$$

$$\text{s.t.} \sum_{a\in\mathcal{A}^b} d_a^b \cdot \pi(w_a^b, \tilde{w}_a^b) \leq B, \tag{13}$$

$$\tilde{\theta}_a^c(\tilde{\mathbf{w}}^b) = \frac{1}{2} \cdot \theta_a^c \quad \text{if } \tilde{w}_a^b \geq 5,\, a \in \mathcal{A}^c \cap \mathcal{A}^b; \tag{14}$$

$\mathbb{L}_{\text{ower}}$:

$$\boldsymbol{\phi}^{b^\star}(\tilde{\mathbf{w}}^b) = \arg\min_{\phi^b \in \mathbb{R}_{\geq 0}^{|\mathcal{A}^b|}} \sum_{a\in\mathcal{A}^b} \bar{d}_a^b(\tilde{\mathbf{w}}^b) \cdot \phi_a^b, \tag{15}$$

$$\text{s.t. } \phi_a^b = \sum_{(u,v)\in\mathcal{V}_{\mathcal{O}}^b \times \mathcal{V}_{\mathcal{D}}^b} q_{(u,v)}^b \cdot \mathbb{I}_{a\in p_{(u,v)}^b} \geq 0,\, \forall a \in \mathcal{A}^b.$$

The *Upper-Level Problem* ($\mathbb{U}_{\text{pper}}$) optimizes the distribution of infrastructure upgrades. The first term in Eq. (12) reduces cyclist exposure by prioritizing links with high flow, traffic, and length. The second term limits additional car travel time due to capacity losses on shared links with DoS $\geq$ 5. The trade-off is managed by the weight $\epsilon \in [0, 1]$. Constraints Eq. (13) and (14) enforce the budget limit and car lane capacity reduction rule. The *Lower-Level Problem* ($\mathbb{L}_{\text{ower}}$) computes bike flows assuming cyclists use shortest paths based on cyclist distance. The resulting flows $\boldsymbol{\phi}^{b^\star}(\tilde{\mathbf{w}}^b)$ influence the upper-level objective via exposure, thus coupling both levels.

Solving Problem 2 is NP-hard as proved in [14]. It generalizes the knapsack problem, where each arc upgrade is a decision with a cost $\pi(w_a^b, \tilde{w}_a^b)$ and coupled benefits for exposure and car flow, making the problem both combinatorial and nonlinear.

III. Solution Algorithm

To solve Problem 2 efficiently, we adopt the genetic algorithm (GA) from [15], which is tailored for bilevel problems. The GA iteratively evolves upgrade configurations

$\tilde{\mathbf{w}}^b$ using selection, crossover, and mutation, evaluating each solution based on cyclist exposure and car travel time under the budget B.

During mutation, solutions exceeding the budget are discarded. To improve budget use, a final optimization step greedily upgrades low-DoS arcs using any remaining budget $\tilde{B}$. While not guaranteeing global optimality, this heuristic provides computationally tractable solutions. Algorithm 1 summarizes the approach. Here, $\Delta_a = \pi(\tilde{w}_a^b, w_{\text{DoS}_6})$ is the cost to upgrade arc a to the highest DoS level.

Algorithm 1 GA

Input: $\tilde{\mathbf{w}}^b$, $\tilde{B}$
Output: Updated arc weights: $\tilde{w}_a^b$, remaining budget: $\tilde{B}$
for $a \in \mathcal{A}^b$ sorted by DoS(a) **do**
  Compute upgrade cost: $\Delta_a = \pi(\tilde{w}_a^b, w_{\text{DoS}_6})$
  if $\tilde{B} \geq \Delta_a$ **then**
    Upgrade arc $\tilde{w}_a^b$ to $w_{\text{DoS}} = 6$.
    Update remaining budget: $\tilde{B} \leftarrow \tilde{B} - \Delta_a$
  end if
  if $\tilde{B} \leq 0$ **then**
    Exit
  end if
end for
Return: Updated $\tilde{w}_a^b$, remaining $\tilde{B}$

IV. Grenoble Network Case Study

In this section, we present a case study based in Grenoble, France, and analyze the optimization results obtained using the GA to solve Problem 2. Additionally, we assess the impact of bike lane upgrades on car travel times using a post-upgrade UE traffic assignment.

A. Problem Parameters and Optimal Solution

To solve the all-or-nothing traffic assignment problem and determine cyclist flows, as described in Section II-C, we use parameters recommended in [23], which better suit the model. Specifically, we set $\rho = 0.04$, $\kappa = 0.14$, and $\psi = 0.84$. The slope factor s_a^b is defined in Table III as a function of the slope m_a^b, where downhill routes are considered as the reciprocal of the uphill slope factor when traversed in the opposite direction.

Slope (m_a^b) %	Uphill Factor (s_a^b)	Downhill Factor (s_a^b)
$m_a < 2$	1.0	1.0
$2 \leq m_a < 4$	1.371	$1/1.371 \approx 0.729$
$4 \leq m_a < 6$	2.203	$1/2.203 \approx 0.454$
$m_a \geq 6$	4.239	$1/4.239 \approx 0.236$

TABLE III: Slope factors for cycling distance calculation

Let $B = \alpha \cdot B_{\max}$ be the available budget, where $B_{\max}$ is the cost to upgrade the full network to DoS 6 and $\alpha \in [0, 1]$. Upgrade costs per meter depend on DoS levels, based on French public estimates [24].

Fig. 2 shows the normalized car flows $\phi^{c^\star}$ obtained via the Frank-Wolfe algorithm [25], [26] for Problem 1, which represents the baseline pre-upgrade traffic assignment. These are used to solve Problem 2 across three budgets $\alpha \in \Phi = \{\frac{1}{3}, \frac{2}{5}, \frac{1}{2}\}$ and eleven ϵ-constraint values

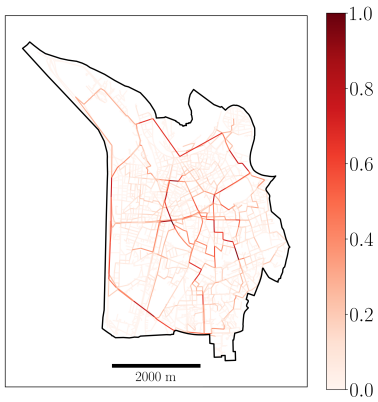


Fig. 2: Original pre-upgrade UE flows (normalized) obtained by solving Problem 1.

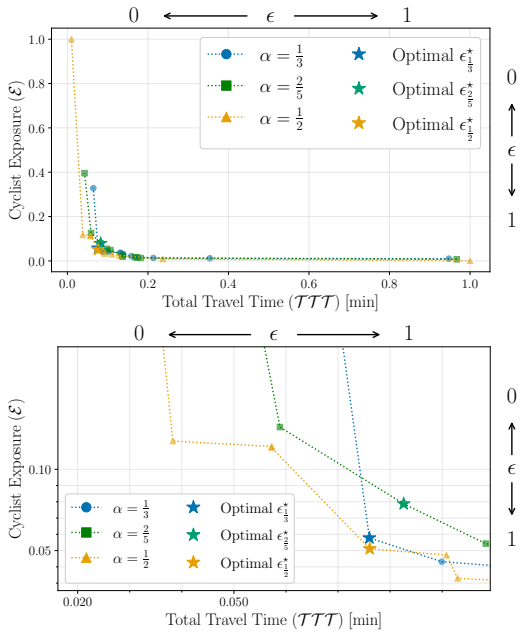
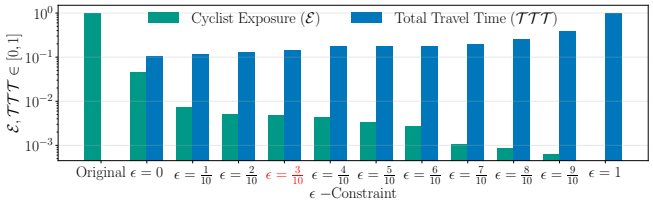


Fig. 3: Normalized travel time versus cyclist exposure. The top plot displays the Pareto fronts resulting from the trade-off Problem 2 across different values of $\epsilon \in \Lambda$ and $\alpha \in \Phi$. The bottom plot presents a zoomed-in view focusing on the optimal ϵ-constraint solutions.

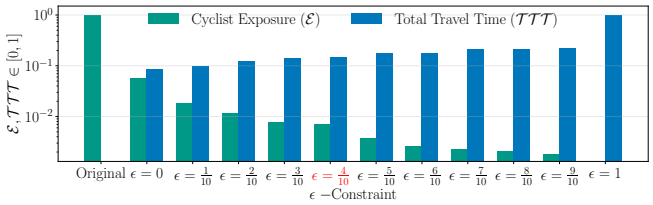
$\epsilon \in \Lambda = \{0, \frac{1}{10}, \ldots, 1\}$. For each α, we evaluate the Pareto set $\{(\epsilon \cdot \mathcal{E}, (1-\epsilon) \cdot \mathcal{TTT})\}_{\epsilon \in \Lambda}$ and define the optimal trade-off as the closest point to the origin:

$$\epsilon_\alpha^\star = \arg\min_{\epsilon \in \Lambda} \sqrt{(\epsilon \cdot \mathcal{E})^2 + ((1-\epsilon) \cdot \mathcal{TTT})^2}. \tag{16}$$

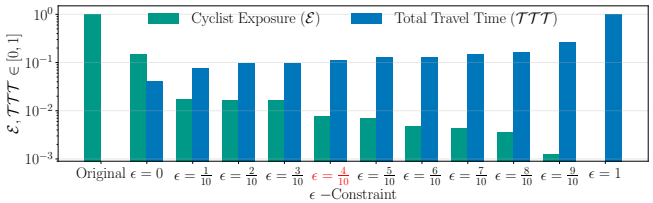
Fig. 3 displays these trade-offs. When $\epsilon = 0$, travel time is minimized; when $\epsilon = 1$, cyclist exposure is prioritized. Higher budgets (e.g., $\alpha = \frac{1}{2}$) yield lower exposure but slightly higher travel times. The optimal trade-offs occur at $\epsilon_{\frac{1}{3}}^\star = \frac{3}{10}$ and $\epsilon_{\frac{2}{5}}^\star = \epsilon_{\frac{1}{2}}^\star = \frac{4}{10}$. Fig. 4 shows normalized results for each ϵ, confirming that as ϵ increases, exposure decreases and travel time rises. Optimal $\epsilon_\alpha^\star$ points are marked in red.



(a) $\alpha = \frac{1}{3}$



(b) $\alpha = \frac{2}{5}$



(c) $\alpha = \frac{1}{2}$

Fig. 4: Normalized cyclist exposure and travel time for each $\epsilon \in \Lambda$ and $\alpha \in \Phi$. In red color are the $\epsilon^\star_{\frac{1}{3}}$, $\epsilon^\star_{\frac{2}{5}}$ and $\epsilon^\star_{\frac{1}{2}}$ from top to bottom, respectively.

B. Impact on Car Routing Choice and Travel Time

To assess the impact of upgrades at $\epsilon^\star_\alpha$, we compare the UE traffic assignment computed on the pre-upgrade network (i.e., assuming drivers maintain their original routes) with the post-upgrade UE assignment, where drivers re-optimize their routes in response to updated capacities. Using the Frank-Wolfe method, Fig. 5 shows that even after route re-optimization, travel times increase only slightly (maximum 11.1% for $\alpha = \frac{1}{2}$), indicating limited disruption to car traffic.

Fig. 6 (a)–(c) shows bike lane improvements focus on high-impact segments, especially with lower budgets. Higher budgets allow for greater, but less continuous expansion. Car flows in Fig. 6 (d)–(f) remain similar to the original (Fig. 2), though some central segments show notable flow changes (e.g., Fig. 6 (f)). These localized redistributions reflect moderate adjustments in user equilibrium, with most flow changes between 20–30%. The broader dispersion under $\alpha = \frac{1}{2}$ aligns with its more extensive upgrades. Overall, these shifts result in modest travel time increases, confirming that targeted upgrades can enhance cyclist safety with minimal impact on vehicular efficiency.

V. Conclusion and Future Work

This work presents a bi-objective optimization framework for designing safer and more efficient cycling infrastructure, demonstrated on Grenoble, France. The problem balances minimizing cyclist exposure and preserving car travel times,

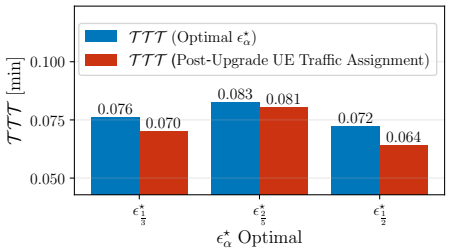


Fig. 5: Comparison of car travel times at the optimal trade-off points $\epsilon^\star_\alpha$, with post-upgrade UE traffic assignment after bike lane upgrades at these locations.

solved using a genetic algorithm (GA). Across varying budgets, the method identifies optimal trade-offs that significantly reduce exposure with minimal impact on car travel times, even after re-routing is accounted for through post-upgrade UE traffic assignment.

The framework offers a practical tool for planners, showing that safety and efficiency can coexist, and modest changes may promote a shift toward cycling. Future extensions could explore behavioral responses, equity, broader multi-modal systems, and alternative optimization approaches.

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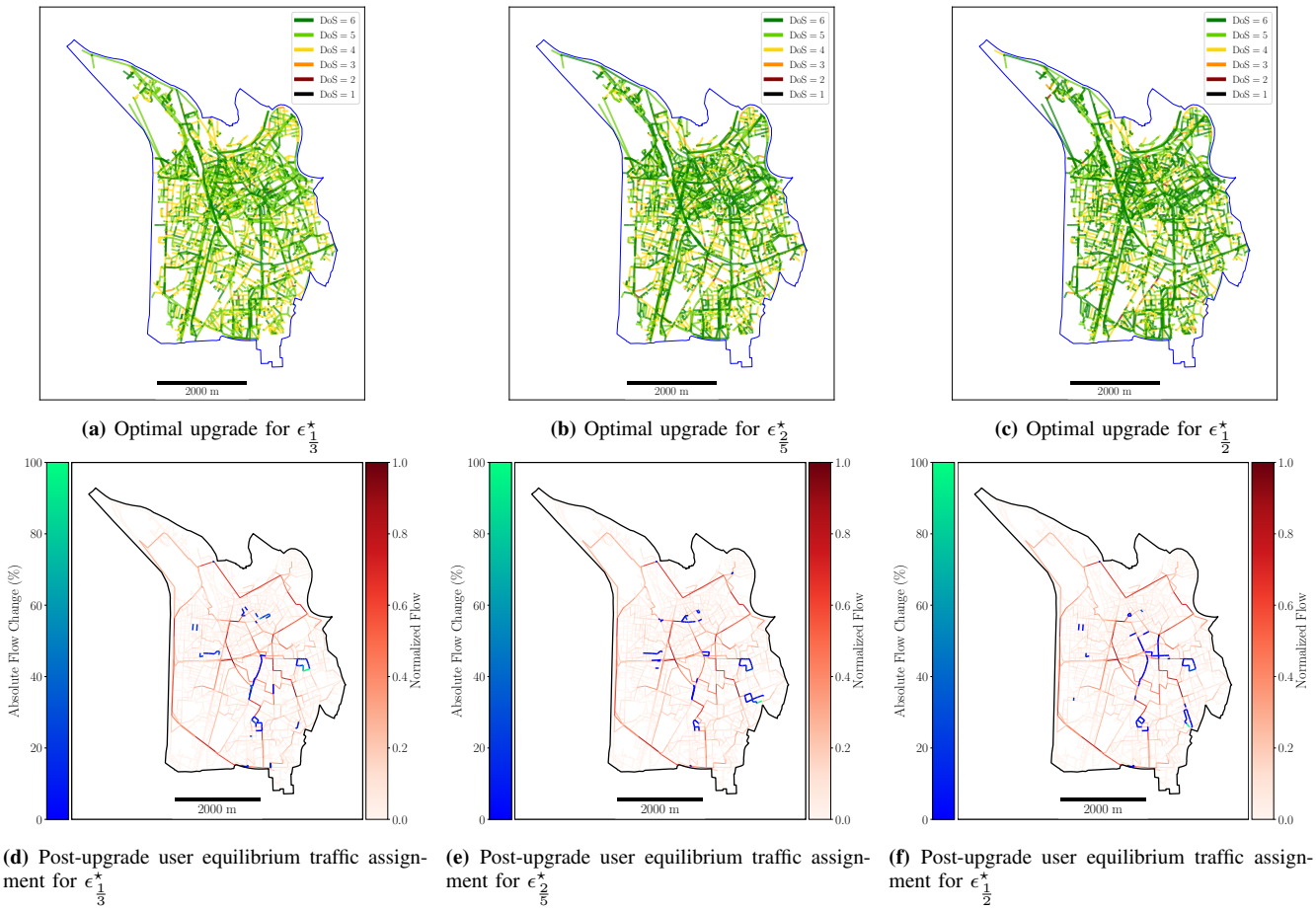
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(a) Optimal upgrade for $\epsilon^\star_{\frac{1}{3}}$

(b) Optimal upgrade for $\epsilon^\star_{\frac{2}{5}}$

(c) Optimal upgrade for $\epsilon^\star_{\frac{1}{2}}$

(d) Post-upgrade user equilibrium traffic assignment for $\epsilon^\star_{\frac{1}{3}}$

(e) Post-upgrade user equilibrium traffic assignment for $\epsilon^\star_{\frac{2}{5}}$

(f) Post-upgrade user equilibrium traffic assignment for $\epsilon^\star_{\frac{1}{2}}$

Fig. 6: The top row shows the optimal bike lane upgrades obtained for $\epsilon^\star_\alpha$. The bottom row displays the post-upgrade UE assignment, where the left color bar highlights the percentage of the absolute flow change only for lanes with the most significant variations, while the right color bar shows the normalized flow distribution across the full network.

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